Received: 13 February 2024 | Accepted: 14 December 2024

DOI: 10.1111/inr.13091

ORIGINAL ARTICLE

International Nursing Review | ICN International Council of Nurses | WILEY

Associations between nurse-to-patient ratio, nurse educational level, and nurse-sensitive patient outcomes: A 12-month prospective observational study

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Funding information
General Research Fund, University Grants Committee, HKSAR, Grant/Award Number: 14102314

Abstract

Aim: To investigate the associations between nurse staffing levels, nurse educational level, and nurse-sensitive patient outcomes among patients in medical and surgical wards.

Background: Patient outcomes are affected by a variety of factors, including nurse staffing and registered nurse (RN) educational levels. An examination of the associations between these factors and patient outcomes will help identify the impact that nurses make on patient care, including health and safety.

Methods: A 12-month prospective observational study was conducted in four major hospitals in Hong Kong. Data on nurse staffing were collected daily, while data on nurse-sensitive patient outcomes were retrieved from patient records and nursing documentation. Multiple regression was used to examine the association between each of the outcomes and the explanatory variables including nurse-to-patient ratio and educational qualifications at the ward level. The STROBE guidelines were followed in this study.

Results: The average monthly nurse-to-patient ratio per ward was 1:9.2 (1:7.4 for day (morning and afternoon) and 1:19.1 for night shifts). The percentage of RNs with a bachelor's or higher degree was 83.6%. A higher nurse-to-patient ratio was significantly associated with a larger number of patients with physical restraints and catheter-associated urinary tract infections. A significant association was also found between a higher percentage of RNs with a bachelor's or higher degree and a smaller number of physical restraints used.

Conclusions: Nurse staffing and educational levels were associated with key patient outcomes such as physical restraint use and incidence of catheter-associated urinary tract infections.

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Implications for nursing and health policy: Current nurse staffing strategies should be refined in light of these findings in order to enhance patient health and safety outcomes.

KEYWORDS
Catheter-associated urinary tract infections, longitudinal study, nurse education, nurse staffing, nurse-sensitive patient outcomes, nurse-to-patient ratio, physical restraints

BACKGROUND

As the largest group of healthcare professionals, nurses have the most direct contact with patients and exert a significant influence on patient outcomes. In particular, evidence suggests that factors such as registered nurse (RN) staffing and educational levels affect a variety of key outcomes including inpatient mortality, falls, medication errors, hospital-acquired infections, pressure injuries, and more (Aiken et al., 2021; Djukic et al., 2019; Musy et al., 2021). However, due to the rising demand for healthcare services caused by factors such as ageing populations, a rise in chronic diseases, and emerging and re-emerging infectious diseases, nurse shortages have become the global norm (Turale & Nantsupawat, 2021). Ensuring an adequate supply of RNs is necessary to enhance healthcare service quality and safety and improve patient outcomes (Fagerström et al., 2018).

In Hong Kong, improving healthcare service quality and retaining staff are among the annual strategic goals of the Hospital Authority, which operates most of the inpatient care services (Hospital Authority, 2021). With only 8.2 nurses per 1,000 people in Hong Kong compared with an average of 10.6 for high-income countries, nurse staffing is suboptimal for a place that also has one of the fastest ageing populations worldwide (Lam, 2023; Wong & Yeung, 2019; World Bank, 2022). The nurse-to-patient ratio also reflects a similar situation, with average ratios of 1:11, 1:12, and 1:24 for the morning, afternoon, and night shifts, respectively across 49 public hospitals, reflecting a large burden on Hong Kong nurses (Lam, 2023). Job dissatisfaction and psychological stress caused by heavy workloads are thus regularly cited as reasons for a high nurse turnover (Hung & Lam, 2020). Despite considerable restructuring and organisational efforts to augment the nursing workforce, such as developing and diversifying nurse education pathways with the expansion of university-based nursing programmes in the past two decades, improving career progression opportunities through the creation of new staff grades such as nurse consultant, and providing competitive remuneration packages, it has remained challenging to maintain a sufficient and optimally proportioned nursing workforce (Ching et al., 2020).

In order to formulate effective nurse staffing policies to meet rising healthcare demand, it is vital to examine the associations between nurse staffing factors and the health and safety of hospitalised patients. Nurse-sensitive patient outcomes (also known as nurse-sensitive quality indicators) are effective in identifying the value of nurse contributions to patient care. Examples of these outcomes include inpatient falls, hospital-acquired infections, pressure injuries, medication errors, and physical restraint use, all of which are largely dependent on the quantity and quality of nursing care received. In addition to staffing, it is necessary to investigate the associations between nurses' educational qualifications and patient outcomes, given that a higher proportion of nurses with a bachelor's or higher degree are linked with reductions in adverse outcomes such as shorter lengths of stay and fewer readmissions (Lasater et al., 2021a). Although numerous studies have been conducted on the associations between these nursing factors and nurse-sensitive outcomes, most have been conducted in Western countries and used a cross-sectional design (Driscoll et al., 2018; Stalpers et al., 2015). Moreover, to our knowledge, no studies have yet been conducted in Hong Kong, where findings may differ from studies in other regions due to variations in healthcare policies, organisational structure, and work culture. Therefore, the aim of this study was to identify and understand the interrelations between nurse staffing, nurse educational level, and nurse-sensitive patient outcomes in Hong Kong.

METHODS

Study design and setting

A prospective observational study was conducted from January 2016 to December 2018 in four major public hospitals offering 24-hour emergency services. A total of 71 wards were recruited for the study, including adult general medical and surgical wards. These wards had 36 to 43 beds and a bed occupancy rate of over 105% during the study period. The protocol for this study has been published and no significant deviations were made to the methodology (Chau et al., 2015). We have adhered to the STROBE guidelines in the report of this study.

Sample

Demographic data were collected every three months over a 12-month period from all full-time nurses in the study wards with direct patient care duties for 50% or more of their job responsibilities. These duties comprise patient observation and assessment, health education, communication, and care coordination and planning. Demographic information such as age, gender, current job position, years of experience, educational attainment at entry to the profession, and highest educational attainment were recorded. Nurses' educational

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attainment at different time points was recorded to reflect professional and skill progression during their careers.

Measures and source of data

Nurse staffing

Information on nurse staffing was collected prospectively from wards, and the central nursing division and human resources departments of the hospitals, including:

- RN staffing per unit-shift (morning, afternoon, and night shifts)
- Number of RNs with a bachelor or higher degree upon entry to the profession staffing per unit-shift
- Nursing care hours per patient day: number of RNs per patient day and number of active nursing staff hours (including RNs and enrolled nurses) per patient day

Nursing care hours per patient day were used to determine eligibility for inclusion in the nurse-to-patient ratio calculation, and senior nurses who spent more time overseeing administrative tasks supporting patient care, coordinating staffing, and mentoring juniors at the time of data collection were excluded from the analysis. Active hours refer to the number of hours worked by nurses with direct patient care responsibilities and exclude paid leaves for illness, vacation, or further education. As the classifications of nurse roles may vary globally, in this study, enrolled nurses refer to those who have completed at least a 2-year diploma programme in nursing while RNs refer to those who have completed a basic general nursing programme of at least 3 years. Enrolled nurses contribute significantly to patient care and typically work under the guidance of RNs.

Nurse-sensitive patient outcomes

Data on nurse-sensitive patient outcomes were retrieved from paper-based patient records, nursing documentation, and through the Clinical Management System (CMS) and Clinical Data Analysis and Reporting System (CDARS) for every shift during the study period. The following outcomes were included:

- Prevalence of hospital-acquired pressure injuries, inpatient falls, physical restraint use, and hospital-acquired catheter-associated urinary tract infections (CAUTIs).

The implementation guide for the National Quality Forum (NQF)-endorsed nursing-sensitive care performance measures was used to develop the definitions and criteria, data sources, and data collection procedures for each outcome (Joint Commission, 2009). These outcomes were selected as they have been found to produce consistent results regarding the influence of nurse staffing on patient health outcomes (Oner et al., 2021).

Data collection procedures

A pilot testing phase was conducted in one medical and one surgical ward for two months to identify potential difficulties in the data collection process and to organise the logistics of the study. Modifications to the data collection process and questionnaires were made as required. Since it was noted that some data on nurse-sensitive patient outcomes such as CAUTIs and physical restraint use were not recorded in the CMS or CDARS, we reached agreements with the relevant hospitals and established pragmatic new routines with frontline nurse colleagues to record the relevant data. As a result, our research team was responsible for charting the daily use of urinary catheters and restraint devices in the surgical wards. Moreover, as data such as the demographic and clinical details of patients diagnosed with CAUTIs were also not recorded in a way suitable for data retrieval by our research nurse, the data had to be retrieved with the support of an infection control nurse through manual screening of large records of specimen results and patient medical records. In addition, we also retrieved fall incidence from paper-based daily nurse shift reports. The research nurse also underwent two training sessions to ensure consistent understanding and to reduce potential errors in data abstraction and coding. Three undergraduate nursing students were also recruited and trained to facilitate data collection. Additionally, an audit on a random sample of 5% of the collected data was conducted by two co-investigators to ensure data accuracy.

Statistical analyses

Data were summarised and presented using appropriate descriptive statistics. The normality of variables with continuous data was assessed using skewness statistics and normal probability plots. The nurse-sensitive patient outcomes of interest in the study were pressure injuries, fall incidence, physical restraint use, and CAUTIs. Average monthly data of the study outcomes collected over 12 months in each participating ward were calculated to quantify their average levels spanning the 12-month study period.

The explanatory variables of interest of the study included the nurse-to-patient ratio, the proportion of RNs with a bachelor or higher degree upon entry to the profession, and the proportion of RNs with a bachelor or higher degree per ward. The average monthly number of patients and RNs in the day (morning and afternoon) and night shifts of duty per ward were used to derive the monthly average nurse-to-patient ratio in each ward. The tri-monthly surveys in the participating wards were used to estimate the proportion of RNs with a bachelor or higher degree upon entry to the profession per ward.

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Multiple regression was used to examine the association between each of the outcomes and each of the explanatory variables at the ward level. Specifically, the average monthly data collected in the participating wards were entered into the regressions. As the study wards were recruited from four hospitals, adjustment was made in the regression analyses to account for the heterogeneity of hospital effect, if any, on the outcomes. The effect of the hospitals was adjusted as a fixed factor in each of the multiple regression models for each of the outcomes and explanatory variables concerned by including three dummy variables with an arbitrarily chosen hospital as a reference. The outcomes of the number of patients with pressure injuries, the number of fall incidence, and the number of patients with CAUTI were all square root transformed before being entered into the regression models to render their residuals normally distributed. All statistical analyses were performed using IBM SPSS 26.0 (IBM Crop., Armonk, NY) with a level of significance set at 0.05 (2-tailed).

Ethical approval

Ethics approval was received from the Joint Chinese University of Hong Kong-New Territories East Cluster Clinical Research Ethics Committee and the Kowloon West Cluster Research Ethics Committee (CRE-2013.101). All methods were performed in accordance with local laws, the Declaration of Helsinki, institutional policies, and the ICH-GCP. We were also compliant with the Hong Kong Personal Data (Privacy) Ordinance. Following an explanation of the study purpose, eligible nurses were invited to participate in the study, and signed informed consent forms were obtained. All questionnaires were conducted anonymously. Approval was obtained to access staffing information, paper-based patient records, nursing documentation, and data from the CMS and CDARS of participating hospitals. All patient identifying information was erased with correction tape. A reference number was assigned to each hospital record and attached to the corresponding data. Data were analysed anonymously.

RESULTS

Participants' characteristics

The number of nurses who participated in each of the four waves of the study ranged from 604 to 865. Most were female (> 70%), aged 55 years or below (> 90%), and had the job title RN (> 80%), while the remainder were advanced practice nurses (13%), nurse specialists, nursing officers, ward managers, or nurse consultants (4%). Advanced practice nurses and nurse specialists possess clinical competencies for advanced, expanded, and specialty practice. Among their responsibilities are supervisory work, planning of care in specific areas of practice, ensuring the delivery of evidence-based care, and mentoring nursing students and juniors. Ward managers and nursing officers oversee departmental/unit administration, coordinate staffing and resource planning, and develop systems to promote safe and effective care delivery. Nurse consultants are responsible for establishing expertise in practice, developing nursing standards and care protocols, leading clinical/research efforts to promote professional nursing practice, and providing nurse-led care (Nursing Council of Hong Kong, 2020). The median length of working experience in the study units ranged from 5.0 to 5.7 years. Most nurses (> 70%) spent an average of 76%–100% of their working time on direct patient care. Half of the participants had a bachelor or higher degree upon entry into the nursing profession and most (> 85%) had a bachelor or higher degree at the time of the study. Further details of participant characteristics are shown in Table 1.

Associations between nurse staffing level and nurse-sensitive patient outcomes

The average monthly nurse-to-patient ratio per ward ($n = 71$) during morning, afternoon, and night shifts was 1:9.2 (SD 2.5) (1:7.4 (SD 2.0) for day (morning and afternoon) and 1:19.1 (SD 7.0) for night shifts) (Table 2). The average monthly number of patients with pressure injuries per ward was 0.63 (SD 0.79), fall incidence per ward was 0.59 (SD 0.5), physical restraint use per ward was 119.2 (SD 72.7), and patients with CAUTI per ward was 0.89 (SD 1.3) (Table 2). Of note, on average, an increasing incidence of physical restraint use and CAUTIs was associated with an increasing level of nurse-to-patient ratio (Supporting Information Table S1). The multiple regression analysis show that a higher nurse-to-patient ratio is significantly associated with a larger number of patients with physical restraints (regression coefficient, B:13.15, 95% CI, 6.89–19.40, $p < 0.001$) and with CAUTIs (square root-transformed B: 0.039, 95% CI, 0.001–0.078, $p = 0.043$) (Table 3).

Associations between nurse educational level and nurse-sensitive patient outcomes

The average monthly percentage of RNs with a bachelor or higher degree per ward was 83.6, (SD 11.1) (Table 2). A higher percentage of RNs with a bachelor or higher degree is significantly associated with a smaller number of physical restraints used (B: −1.98, 95% CI, −3.39 to −0.57, $p = 0.007$) (Table 4). No statistically significant associations were found between nurse-sensitive outcomes and the percentage of RNs with a bachelor or higher degree upon entry into the profession (Table 5).

DISCUSSION

This is the first study to investigate the association between nurse factors and nurse-sensitive patient outcomes in Hong Kong. Our findings shed light on the nurse staffing levels in

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TABLE 1 Characteristics of the nurse participants in the surveys.

	Wave of survey			
Characteristics	**First (n = 865)**	**Second (n = 604)**	**Third (n = 680)**	**Fourth (n = 661)**
Sex				
Female	74.3%	74.4%	78.2%	75.9%
Male	25.7%	25.6%	21.8%	24.1%
Age (years)				
≤ 25	21.1%	22.5%	19.9%	22.8%
26–30	31.5%	34.7%	35.8%	37.1%
31–35	11.9%	10.6%	10.5%	10.4%
36–40	10.1%	10.9%	12.7%	10.4%
41–45	6.9%	8.1%	9.1%	8.1%
46–50	5.6%	7.6%	6.9%	6.8%
51–55	3.9%	4.0%	4.2%	3.2%
56–60	1.1%	1.6%	0.9%	1.0%
> 60	7.9%	0.0%	0.0%	0.3%
Current job title				
RN	82.0%	83.3%	82.3%	82.6%
APN	13.0%	12.4%	13.7%	13.8%
NO	1.4%	1.8%	1.3%	1.0%
NS	0.4%	0.7%	0.6%	0.7%
WM	3.1%	1.8%	1.9%	1.8%
NC	0.0%	0.0%	0.2%	0.2%
Average working time on direct patient care				
51%–75%	30.0%	29.3%	29.3%	29.5%
76%–100%	70.0%	70.7%	70.7%	70.5%
Educational level at entry of RN				
Diploma	49.1%	50.5%	50.0%	47.3%
Bachelor (pre-registration)	47.9%	46.5%	46.7%	50.9%
Master (pre-registration)	3.0%	3.0%	3.3%	1.8%
Highest educational attainment				
Diploma	13.3%	14.6%	12.2%	13.4%
Bachelor	60.6%	60.4%	61.8%	60.5%
Master	26.1%	25.0%	25.2%	24.8%
DN	0.0%	0.0%	0.6%	1.3%
PhD	0.0%	0.0%	0.2%	0.0%
Years of experience on current position[a]	4.0 (2.0–8.3)	4.1 (1.7–9.0)	3.9 (1.9–8.0)	3.7 (1.9–7.4)
Years of experience as an registered nurse[a]	5.7 (2.4–14.0)	5.0 (2.0–14.3)	5.4 (2.1–15.0)	5.3 (2.3–14.7)
Working duration in current ward (months)[a]	36 (12–84)	28 (9–79)	33 (12–84)	30 (12–85)

Data of variables marked with [a] are presented as median (inter-quartile range), and all others are presented as %.
Abbreviations: APN, advanced practice nurse; DN, Doctor of Nursing; NC, nurse consultant; NO, nursing officer; PhD, Doctor of Philosophy; RN, registered nurse; NS, nurse specialist; WM, ward manager.

medical and surgical wards of acute hospitals and indicate that patient safety might be improved by refining current nurse staffing strategies. In terms of impact on patient outcomes, every additional patient per nurse was associated with a significantly increased risk of physical restraint use and CAUTIs. Educational qualifications were also found to be associated with physical restraint use, with more highly educated nurses less likely to use restraints.

The average monthly nurse-to-patient ratio of 1:9.2 (SD 2.5) per ward during day and night shifts reported in the

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TABLE 2 Nurse staffing level and nurse-sensitive patient outcomes of the study wards.

	Mean (SD)	Inter-quartile range
Nurse staffing level		
Average monthly number of patients per ward	1175.6 (290.3)	996.6–1397.5
Average monthly number of RNs per ward	133.1 (32.9)	109.3–152.0
Average monthly nurse-to-patient ratio per ward	9.2 (2.5)	8.0–10.2
Average monthly percentage of RNs with a bachelor or higher degree upon entry to the profession per ward	49.7 (16.4)	40.6–62.7
Average monthly percentage of RNs with a bachelor or higher degree per ward	83.6 (11.1)	77.4–91.6
Nurse-sensitive patient outcomes		
Average monthly number of patients with pressure injuries per ward	0.63 (0.79)	0.00–0.75
Average monthly number of fall incidence per ward	0.59 (0.50)	0.25–0.90
Average monthly number of physical restraints per ward	119.2 (72.7)	64.7–164.7
Average monthly number of patients with CAUTIs per ward	0.89 (1.30)	0.00–1.17

Abbreviations: CAUTIs, catheter-associated urinary tract infections; RN, registered nurse.

TABLE 3 Associations between nurse-sensitive patient outcomes and nurse-to-patient ratio.

Nurse-sensitive patient outcomes	Nurse-to-patient ratio	
	B (95% CI)	*p*
Number of patients with pressure injuries[a]	−0.009 (−0.056, 0.038)	0.705
Number of fall incidence[a]	0.019 (−0.022, 0.059)	0.360
Number of physical restraints	13.15 (6.89, 19.40)	<0.001
Number of patients with CAUTIs[a]	0.039 (0.001, 0.078)	0.043

The outcomes marked with [a] were square root transformed before being subjected to multiple regression analysis to render their residuals normally distrusted in the models.
B: Regression coefficient estimated by multiple regression with adjustment for the heterogeneity of hospital effect, if any, on the outcomes by including three dummy variables with an arbitrarily chosen hospital as a reference; it represents the average number of each (transformed) outcome increased when the nurse-to-patient ratio is increased by 1.
Abbreviation: CAUTIs, catheter-associated urinary tract infections.

current study was found to be slightly higher than that of European countries, which had an average ratio of 1:8.3 (SD 2.4) across 300 hospitals (Aiken et al., 2014). However, the ratio was notably higher than in regions such as Australia (ratio: 1:5.52) where policies have been implemented to regulate the maximum number of patients assigned to an RN, and England (1:6.9), which provides national recommendations for nurse staffing (Bridges et al., 2019; McHugh et al., 2020; National Health Service England, 2021). It is concerning to find a relatively high ratio in Hong Kong as nurses responsible for large numbers of patients may experience

TABLE 4 Associations between nurse-sensitive patient outcomes and percentage of registered nurses with a bachelor or higher degree per ward.

Nurse-sensitive patient outcomes	Percentage of registered nurses with a bachelor or higher degree per ward	
	B (95% CI)	*p*
Number of patients with pressure injuries[a]	−0.002 (−0.012, 0.007)	0.621
Number of fall incidence[a]	0.000 (−0.008, 0.009)	0.913
Number of physical restraints	−1.98 (−3.39, −0.57)	0.007
Number of patients with CAUTIs[a]	−0.002 (−0.010, 0.007)	0.704

The outcomes marked with [a] were square root transformed before being subjected to multiple regression analysis to render their residuals normally distrusted in the models.
B: Regression coefficient estimated by multiple regression with adjustment for the heterogeneity of hospital effect, if any, on the outcomes by including three dummy variables with an arbitrarily chosen hospital as a reference; it represents the average number of each (transformed) outcome increased when the percentage of registered nurses with a bachelor or higher degree per ward is increased by 1.
CAUTIs, catheter-associated urinary tract infections.

TABLE 5 Associations between nurse-sensitive patient outcomes and percentage of registered nurses with a bachelor or higher degree upon entry to the profession.

Nurse-sensitive patient outcomes	Percentage of registered nurses with a bachelor or higher degree upon entry into the profession	
	B (95% CI)	*p*
Number of patients with pressure injuries[a]	0.001 (−0.007, 0.010)	0.720
Number of fall incidence[a]	0.003 (−0.004, 0.010)	0.368
Number of physical restraints	−0.06 (−1.29, 1.18)	0.924
Number of patients with CAUTIs[a]	−0.001 (−0.008, 0.006)	0.724

The outcomes marked with [a] were square root transformed before subjected to multiple regression analysis to render their residuals normally distrusted in the models.
B: Regression coefficient estimated by multiple regression with adjustment for the heterogeneity of hospital effect, if any, on the outcomes by including three dummy variables with an arbitrarily chosen hospital as reference; it represents the average number of each (transformed) outcome increased when the percentage of registered nurses with a bachelor or higher degree upon entry to the profession is increased by 1.
Abbreviation: CAUTIs: catheter-associated urinary tract infections.

excess stress and fatigue, thus compromising their ability to maintain an appropriate standard of patient care. In a study among 116 U.S. hospitals, Lasater et al. (2021b) found that each additional patient per nurse was associated with higher odds of in-hospital mortality, 60-day mortality, 60-day readmission, and longer lengths of stay. On the other hand, a recent evaluation of minimum nurse-to-patient ratio legislation in Queensland, Australia, which mandates four patients or less per nurse during morning/afternoon shifts and seven patients or less during night shifts, highlighted that reduced nursing workloads were associated with lower mortality and readmission (McHugh et al., 2021). It is, therefore, recommended that hospitals in Hong Kong reconsider their staffing strategies to ensure that nurses have adequate time to deliver continued quality patient care and enable them to practise to their best ability.

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Another significant finding is that the average number of physical restraints used monthly was significantly associated with a higher nurse-to-patient ratio, increasing more than tenfold with every additional patient per nurse. This is a notable finding given the paucity of studies investigating the association between nurse staffing and physical restraints in general medical and surgical wards despite evidence of their connection in other settings such as critical care units (Driscoll et al., 2018; Old & Cramer, 2021). Though the use of restraints is often justified as a patient safety measure to prevent adverse incidents such as falls, the removal of indwelling medical devices or threats to staff safety (Thomann et al., 2021), it has been associated with a plethora of negative patient experiences and outcomes including a loss of personal autonomy and freedom, lengthier hospital stays, and functional decline (Chou et al., 2020; Strout, 2010). In fact, there is substantial evidence to suggest that restraint use is ineffective in increasing patient safety and alternative methods which preserve patient dignity are recommended to replace restraint use and avoid compounding stressful patient experiences (Sze et al., 2012; Wong et al., 2020). However, the decision to use restraints is often dependent on staff availability, as alternative interventions such as patient observation or communication are unfeasible due to a shortage of staff on duty (Salehi et al., 2021). Consequently, physical restraints continue to be applied according to the norm, and this is reflected in our study. To decrease reliance on physical restraint use and encourage the adoption of patient-friendly methods, it is imperative to bolster nursing staff and ensure adequate workforce supply in order to devise the best possible care plans for patients.

Besides physical restraint use, a higher nurse-to-patient ratio was also associated with a slight increase in the number of patients with a CAUTI, reflecting results from previous studies on the link between nurse staffing and nosocomial infections (Driscoll et al., 2018; Van et al., 2020). Associated with longer lengths of hospital stay and higher inpatient mortality, CAUTIs have a detrimental impact on patient health (Mitchell et al., 2016). Our findings show that there is an average monthly CAUTI incidence of 0.89 (SD: 1.3) per ward and that a higher nurse-to-patient ratio leads to higher monthly CAUTI incidence. This is consistent with results from a recent 9-year study on healthcare-associated infection rates from the United States which found that periods with more nurses coincided with significantly fewer CAUTIs (Clifford et al., 2022). Considering the benefit of having more nurses available to support catheter insertion and management (Belizario, 2015; Fletcher-Gutowski & Cecil, 2019), it seems appropriate for wards to have sufficient nurses to facilitate catheter use and reduce the number of avoidable hospital-acquired CAUTIs.

In contrast to findings from previous studies (Wang et al., 2020), no significant associations were found between nurse-to-patient ratio and outcomes of pressure injuries and falls, which may reflect the effect of mediating factors, such as nursing skill mix or nurse leadership, on these outcomes (Dall'Ora et al., 2022; Lee et al., 2023). A systematic review of 19 studies also indicates that the type of nurse staffing measure may affect the significance of staffing effects on hospital-acquired conditions, with studies using RN hours per patient day finding non-significant impacts on patient outcomes, which may be the case in the present study (Shin et al., 2019). Moreover, with regard to the relationship between patient falls and nurse staffing, a recent retrospective study indicates that a reduced risk of falls was observed in wards with a higher number of patients, whereas no significant association was found with a higher nurse-to-patient ratio, suggesting that the presence of nurses may not have as much an effect on the prevention of falls as the number or characteristics of patients in a ward, who may be able to assist one another and manage fall risks (Seeherunwong et al., 2022).

Besides the number of RNs on a shift, nurses' educational qualifications have also been linked with patient outcomes. As evidence-based practice has become increasingly incorporated into tertiary nursing education, nurses with a bachelor or higher degree exhibit higher levels of clinical reasoning and improved competencies in patient care and safety (Mackey & Bassendowski, 2017; Vaismoradi et al., 2020). Accordingly, studies indicate that a higher proportion of RNs with a bachelor degree is associated with reduced risk of patient mortality and shorter lengths of stay (Lasater et al., 2021a; Porat-Dahlerbruch et al., 2022). Our study adds to this evidence, suggesting that the odds of using physical restraints decrease by nearly two times with every one-point increase in the percentage of RNs with a bachelor or higher degree per ward, demonstrating that nurses with undergraduate or graduate degrees were more likely to find alternatives or forego the use of physical restraints. However, no associations were found with pressure injuries, falls, and CAUTIs, suggesting that RNs' educational level may exert a less significant effect on patient outcomes as compared with other factors such as staffing levels, thus underscoring more complex interactions between nursing factors and patient outcomes.

In a positive finding, over 80% of our RN sample had at least a bachelor degree, which is significantly higher than worldwide averages (Europe: 52%; United States: 56%; South Korea: 63%) and in line with U.S. recommendations for an effective nursing workforce (Aiken et al., 2014; Ball et al., 2018; Shin et al., 2020). Moreover, the average monthly percentage of RNs with a bachelor or higher degree upon entry into the profession was also high at about 50%. These findings reflect the development of nursing education in Hong Kong, which shifted in the early 1990s from a largely hospital-based apprenticeship system to university-based nursing programmes, leading to a higher proportion of university-educated nurses in recent years (Thompson, 2006). Our findings are reassuring as they suggest that nurses in Hong Kong are increasingly qualified and possess the ability to support the implementation of evidence-based practice, thereby potentially improving patient outcomes.

Limitations

This study has some limitations. As nursing data were collected every three months, all patients admitted during the

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following three months were assumed to have been exposed to the same summary ward-level nursing factors, meaning that if there were any changes to nursing factors during the three months, their effect would not be recorded or accounted for in the statistical analyses. Although averages of 12-month aggregated monthly data of the nurse-sensitive patient outcomes, which were anticipated to be robust to the influences of individual patient characteristics and sickness severity and other seasonal factors, were used in the analysis, we did not collect and make adjustment for proper measures of churn and patient acuity parameters, which may threaten the internal validity of the study findings. Caution should also be taken in the interpretation of the findings as the data were collected from 2016 to 2018 and may not fully reflect the current nurse staffing situation in Hong Kong. In addition, Hong Kong has no law mandating nurse staffing levels, making it difficult to compare current statistics. Moreover, since only medical and surgical wards of public acute hospitals were recruited to the study, results may not be generalisable to private or rehabilitation hospitals which may vary in terms of nurse staffing features and their nurse-to-patient ratio.

CONCLUSION

This study is the first to provide evidence of the associations between nurse staffing factors and patient outcomes in public hospitals in Hong Kong. Results show that increases in the nurse-to-patient ratio are linked to the use of more physical restraints and higher CAUTI incidence per ward. On the other hand, a higher proportion of nurses with bachelor's or higher degrees was associated with reduced physical restraint use. Therefore, effective workforce planning strategies and supporting higher nurse education should be revisited in order to optimise patient safety and outcomes in Hong Kong.

Implications for nursing and health policy

Ensuring an adequately staffed and skilled nursing workforce is a critical part of effective health policy. As our findings highlight the associations between nurse staffing and patient outcomes such as increased physical restraint use and CAUTI incidence, it seems timely for nurse leaders to advocate for more appropriate staffing strategies in order to meet increasing healthcare demands and improve the quality of patient care. Besides having a positive effect on patient outcomes, adequate staffing would also alleviate nurses' work burden and improve their overall practice and well-being. Additionally, as our findings also suggest that higher educational qualifications are associated with less use of physical restraints, nurse leaders should encourage their staff to undertake more development opportunities in order to build and expand on their existing skillsets so as to enhance the standard of care provided. Future studies should explore further complex interactions, such as any potential mediatory relationships which may affect the extent to which nurse staffing and educational level influence patient outcomes.

AUTHOR CONTRIBUTIONS

Study design: Janita Pak Chun Chau and Suzanne Hoi Shan Lo. *Data collection*: Janita Pak Chun Chau, Suzanne Hoi Shan Lo, Danny Wah Kun Tong, Angela Mei Ling Kwok, Eric Lu Shek Chan, Iris Fung Kam Lee, and Diana Tze Fan Lee. *Data analysis*: Kai Chow Choi and Matthew D. Mchugh. *Study supervision*: Janita Pak Chun Chau and Suzanne Hoi Shan Lo. *Manuscript writing*: Janita Pak Chun Chau, Suzanne Hoi Shan Lo, Kai Chow Choi, Laveeza Butt, and David R. Thompson. *Critical revisions for important intellectual content*: Janita Pak Chun Chau, Kai Chow Choi, Laveeza Butt, Suzanne Hoi Shan Lo, and David R. Thompson.

ACKNOWLEDGMENTS

This study was supported by the General Research Fund, University Grants Committee, HKSAR (Ref no.: 14102314).

CONFLICT OF INTEREST STATEMENT

No conflict of interest has been declared by the authors.

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REFERENCES

Aiken, L.H., Simonetti, M., Sloane, D.M., Cerón, C., Soto, P., Bravo, D., et al. (2021) Hospital nurse staffing and patient outcomes in Chile: a multilevel cross-sectional study. *The Lancet Global Health*, 9(8), E1145–E1153. https://doi.org/10.1016/S2214-109X(21)00209-6

Aiken, L.H., Sloane, D.M., Bruyneel, L., Van den Heede, K., Griffiths, P., Busse, R., et al. (2014) Nurse staffing and education and hospital mortality in nine European countries: a retrospective observational study. *Lancet*, 383(9931), 1824–1830. https://doi.org/10.1016/S0140-6736(13)62631-8

Ball, J.E., Bruyneel, L., Aiken, L.H., Sermeus, W., Sloane, D.M., Rafferty, A.M., et al. (2018) Peri-operative mortality, missed care and nurse staffing in nine countries: a cross-sectional study. *International Journal of Nursing Studies*, 78, 10–15. https://doi.org/10.1016/j.ijnurstu.2017.08.004

Belizario, S. (2015) Preventing urinary tract infections with a two-person catheter insertion procedure. *Nursing*, 45(3), 67–69. https://doi.org/10.1097/01.NURSE.0000460736.74021.69

Bridges, J., Griffiths, P., Oliver, E., Pickering, R.M. (2019) Hospital nurse staffing and staff-patient interactions: an observational study. *BMJ Quality & Safety*, 28(9), 706–713. https://doi.org/10.1136/bmjqs-2018-008948

14667657, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/inr.13091 by University Of Pennsylvania, Wiley Online Library on [13/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Chau, J.P.C., Lo, S.H.L., Choi, K.C., Chan, E.L.S., McHugh, M.D., Tong, D.W.K., et al. (2015) A longitudinal examination of the association between nurse staffing levels, the practice environment and nurse-sensitive patient outcomes in hospitals. *BMC Health Services Research*,15, 538. https://doi.org/10.1186/s12913-015-1198-0

Ching, S. S., Cheung, K. & Mak, Y. W. (2020). The development and current challenges of nursing education in Hong Kong. In *Routledge international handbook of nurse education* (1st edn., pp. 50–63). Routledge. https://doi.org/10.4324/9781351121675-5

Chou, M.-Y., Hsu, Y.-H., Wang, Y.-C., Chu, C.-S., Liang, C., Chen, L.-K., et al. (2020) The adverse effects of physical restraint use among older adult patients admitted to the internal medicine wards: a hospital-based retrospective cohort study. *Journal of Nutrition, Health & Aging*, 24, 160–165. https://doi.org/10.1007/s12603-019-1306-7

Clifford, R. J., Newhart, D., Laguio-Vila, M. R., Gutowski, J. L., Bronstein, M. Z. & Lesho, E. P. (2022) Infection preventionist staffing levels and rates of 10 types of healthcare-associated infections: a 9-year ambidirectional observation. *Infection Control and Hospital Epidemiology*, 43(11), 1641–1646. https://doi.org/10.1017/ice.2021.507

Dall'Ora, C., Saville, C., Rubbo, B., Turner, L., Jones, J. & Griffiths, P. (2022) Nurse staffing levels and patient outcomes: a systematic review of longitudinal studies. *International Journal of Nursing Studies*, 134, 104311–104311. https://doi.org/10.1016/j.ijnurstu.2022.104311

Djukic, M., Stimpfel, A.W. & Kovner, C. (2019) Bachelor's degree nurse graduates report better quality and safety educational preparedness than associate degree graduates. *The Joint Commission Journal on Quality and Patient Safety*, 45(3), 180–186. https://doi.org/10.1016/j.jcjq.2018.08.008

Driscoll, A., Grant, M.J., Carroll, D., Dalton, S., Deaton, C., Jones, I., et al. (2018) The effect of nurse-to-patient ratios on nurse-sensitive patient outcomes in acute specialist units: a systematic review and meta-analysis. *European Journal of Cardiovascular Nursing*, 17(1), 6–22. https://doi.org/10.1177/1474515117721561

Fagerström, L., Kinnunen, M. & Saarela, J. (2018) Nursing workload, patient safety incidents and mortality: an observational study from Finland. *BMJ Open*, 8, e016367. https://doi.org/10.1136/bmjopen-2017-016367

Fletcher-Gutowski, S. & Cecil, J. (2019) Is 2-person urinary catheter insertion effective in reducing CAUTI? *American Journal of Infection Control*, 47(12), 1508–1509. https://doi.org/10.1016/j.ajic.2019.05.014

Hospital Authority. *Hospital Authority annual report 2020–2021*. Available at: https://www.ha.org.hk/haho/ho/cc/HA_Annual_Report_2020-21_en.pdf [Accessed 12th June 2023].

Hung, M.S.Y., Lam, S.K.K. (2020) Antecedents and contextual factors affecting occupational turnover among registered nurses in public hospitals in Hong Kong: a qualitative descriptive study. *International Journal of Environmental Research and Public Health*, 17(11), 3834. https://doi.org/10.3390/ijerph17113834

Joint Commission. (2009) *The implementation guide for the NQF endorsed nursing-sensitive care measure set*. Available at: https://docplayer.net/12842065-Implementation-guide-for-the-nqf-endorsed-nursing-sensitive-care-measure-set.html [Accessed 16th January 2024].

Lam, G. (2023) Shortage of nurses in Hong Kong: the challenges ahead. *Asian Education and Development Studies*, 12, 89–102. https://doi.org/10.1108/AEDS-08-2021-0179

Lasater, K.B., Sloane, D.M., McHugh, M.D., Porat-Dahlerbruch, J. & Aiken, L.H. (2021a) Changes in proportion of bachelor's nurses associated with improvements in patient outcomes. *Research in Nursing & Health*, 44(5), 787–795. https://doi.org/10.1002/nur.22163

Lasater, K.B., Sloane, D.M., McHugh, M.D., Cimiotti, J.P., Riman, K.A., Martin, B., Alexander, M. & Aiken, L. H. (2021b) Evaluation of hospital nurse-to-patient staffing ratios and sepsis bundles on patient outcomes. *American Journal of Infection Control*, 49(7), 868–873. https://doi.org/10.1016/j.ajic.2020.12.002

Lee, S. E., Hyunjie, L. & Sang, S. (2023) Nurse managers' leadership, patient safety, and quality of care: a systematic review. *Western Journal of Nursing Research*, 5(2), 176–185. https://doi.org/10.1177/01939459221114079

Mackey, A. & Bassendowski, S. (2017) The history of evidence-based practice in nursing education and practice. *Journal of Professional Nursing*, 33(1), 51–55. https://doi.org/10.1016/j.profnurs.2016.05.009

McHugh, M.D., Aiken, L.H., Windsor, C., Douglas, C. & Yates, P. (2020) Case for hospital nurse-to-patient ratio legislation in Queensland, Australia, hospitals: an observational study. *BMJ Open*, 10(9), e036264. https://doi.org/10.1136/bmjopen-2019-036264

McHugh, M.D., Aiken, L.H., Sloane, D.M., Windsor, C., Douglas, C., Yates, P. (2021) Effects of nurse-to-patient ratio legislation on nurse staffing and patient mortality, readmissions, and length of stay: a prospective study in a panel of hospitals. *Lancet*, 397(10288), 1905–1913. https://doi.org/10.1016/S0140-6736(21)00768-6

Mitchell, B.G, Ferguson, J.K., Anderson, M., Sear, J. & Barnett, A. (2016) Length of stay and mortality associated with healthcare-associated urinary tract infections: a multi-state model. *Journal of Hospital Infection*, 93(1), 92–99. https://doi.org/10.1016/j.jhin.2016.01.012

Musy, S.N., Endrich, O., Leichtle, A.B., Griffiths, P., Nakas, C.T. & Simon, M. (2021) The association between nurse staffing and inpatient mortality: a shift-level retrospective longitudinal study. *International Journal of Nursing Studies*, 120, 103950. https://doi.org/10.1016/j.ijnurstu.2021.103950

National Health Service England. (2021) *Safer staffing: nursing and midwifery*. NHS England. https://www.england.nhs.uk/nursingmidwifery/safer-staffing-nursing-and-midwifery/ [Accessed 20th July 2024].

Nursing Council of Hong Kong (2020) *Scope of core-competences of an advanced practice nurse*. Available at: https://www.nchk.org.hk/filemanager/en/pdf/Scope_of_Core_Competencies_of_an_APN_eng.pdf [Accessed 29th July 2024].

Olds, D. & Cramer, E. (2021) Predictors of physical restraint use on critical care units: an observational structural equation modeling approach. *International Journal of Nursing Studies*, 118, 103925.

Oner, B., Zengul, F.D., Oner, N., Ivankova, N.V., Karadag, A. & Patrician, P.A. (2021). Nursing-sensitive indicators for nursing care: a systematic review (1997–2017). *Nursing Open*, 8(3), 1005–1022. https://doi.org/10.1002/nop2.654

Porat-Dahlerbruch, J., Aiken, L.H., Lasater, K.B., Sloane, D.M. & McHugh, M.D. (2022). Variations in nursing baccalaureate education and 30-day inpatient surgical mortality. *Nursing Outlook*, 70(2), 300–308. https://doi.org/10.1016/j.outlook.2021.09.009

Salehi, Z., Joolaee, S., Hajibabaee, F. & Ghezeljeh, T.N. (2021) The challenges of using physical restraint in intensive care units in Iran: a qualitative study. *Journal of the Intensive Care Society*, 22(1), 34–40. https://doi.org/10.1177/1751143719892785

Seeherunwong, A., Thunyadee, C., Vanishakije, W. et al. (2022) Staffing and patient-related factors affecting inpatient falls in a psychiatric hospital: a 5-year retrospective matched case–control study. *International Journal of Mental Health Systems*, 16, 3. https://doi.org/10.1186/s13033-022-00514-1

Shin, S., Oh, S.J., Kim, J., Lee, I. & Bae, S. (2020) Impact of nurse staffing on intent to leave, job satisfaction, and occupational injuries in Korean hospitals: a cross-sectional study. *Nursing & Health Sciences*, 22(3), 658–666. https://doi.org/10.1111/nhs.12709

Shin, S., Park, J. & Bae, S. (2019). Nurse staffing and hospital-acquired conditions: a systematic review. *Journal of Clinical Nursing*, 28(23–24), 4264–4275. https://doi.org/10.1111/jocn.15046

Stalpers, D., de Brouwer, B.J.M., Kaljouw, M.J. & Schuurmans, M.J. (2015) Associations between characteristics of the nurse work environment and five nurse-sensitive patient outcomes in hospitals: a systematic review of literature. *International Journal of Nursing Studies*, 52(4), 817–832. https://doi.org/10.1016/j.ijnurstu.2015.01.005

Strout, T.D. (2010) Perspectives on the experience of being physically restrained: an integrative review of the qualitative literature. *International Journal of Mental Health Nursing*, 19(6), 416–427. https://doi.org/10.1111/j.1447-0349.2010.00694.x

Sze, T.W., Leng, C.Y. & Lin, S.K.S. (2012) The effectiveness of physical restraints in reducing falls among adults in acute care hospitals and nursing homes: a systematic review. *JBI Library of Systematic Reviews*, 10(5), 307–351. https://doi.org/10.11124/jbisrir-2012

Thomann, S., Zwakhalen, S., Richter, D., Bauer, S. & Hahn, S. (2021). Restraint use in the acute-care hospital setting: a cross-sectional multi-centre study.

14667657, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/inr.13091 by University Of Pennsylvania, Wiley Online Library on [13/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

International Journal of Nursing Studies, 114, 103807. https://doi.org/10.1016/j.ijnurstu.2020.103807

Thompson, D.R. (2006) Nursing in Hong Kong: issues and challenges. *Nursing Science Quarterly*, 19(2), 158–162. https://doi.org/10.1177/0894318406286834

Turale, S. & Nantsupawat, A. (2021) Clinician mental health, nursing shortages and the COVID-19 pandemic: crises within crises. *International Nursing Review*, 68(1), 12–14. https://doi.org/10.1111/inr.12674

Vaismoradi, M., Tella, S., Logan, P.A., Khakurel, J. & Vizcaya-Moreno, F. (2020) Nurses' adherence to patient safety principles: a systematic review. *International Journal of Environmental Research and Public Health*, 17(6), 2028. https://doi.org/10.3390/ijerph17062028

Van, T., Annis, A.M., Yosef, F., Robinson, C.H., Duffy, S.A., Li, Y., et al. (2020) Nurse staffing and healthcare-associated infections in a national healthcare system that implemented a nurse staffing directive: multi-level interrupted time series analyses. *International Journal of Nursing Studies*, 104, 103531. https://doi.org/10.1016/j.ijnurstu.2020.103531

Wang, L., Lu, H., Dong, X., Huang, X., Li, B., Wan, Q., et al. (2020) The effect of nurse staffing on patient-safety outcomes: a cross-sectional survey. *Journal of Nursing Management*, 28(7), 1758–1766. https://doi.org/10.1111/jonm.13138

Wong, A.H., Ray, J.M., Rosenberg, A., Crispino, L., Parker, J. McVaney, C., et al. (2020) Experiences of individuals who were physically restrained in the emergency department. *JAMA Network Open*, 3(1), e1919381. https://doi.org/10.1001/jamanetworkopen.2019.19381

Wong, K., Yeung, M. (2019) *Population ageing trend of Hong Kong*. Office of the Government Economist, Hong Kong. Available at: https://www.hkeconomy.gov.hk/en/pdf/el/el-2019-02.pdf [Accessed 16th January 2024].

World Bank. (2024) *Nurses and midwives (per 1,000 people): high income*. Available at: https://data.worldbank.org/indicator/SH.MED.NUMW.P3?contextual=default&end=2019&locations=XD&most_recent_value_desc=true&start=2000 [Accessed 10th January 2024].

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How to cite this article: Chau, J.P.C., Lo, S.H.S., Choi, K.C., Tong, D.W.K., Kwok, A.M.L., Butt, L. et al. (2025) Associations between nurse-to-patient ratio, nurse educational level, and nurse-sensitive patient outcomes: a 12-month prospective observational study. *International Nursing Review*, 72, 1–10. https://doi.org/10.1111/inr.13091

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